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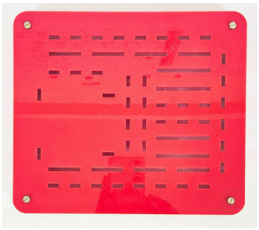
SESSION NAME

Session

5

STRUCTURE AND FASTENERS OF MICROCONTROLLERS

1



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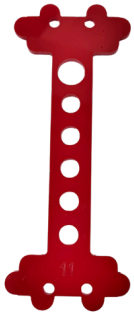


1

Spacers

2

L Clamp

3

Spacer Screws

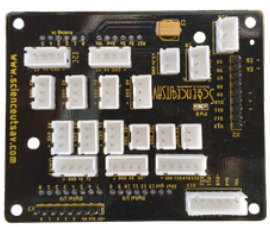
4

Nuts

5

Screw

6

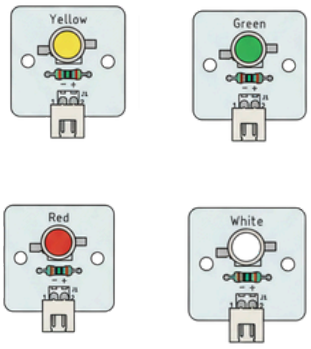


Arduino With Shield

7

Arduino cable

ELECTRONIC COMPONENTS



1 Watt LED PCB

1 Watt LED is a high-brightness light-emitting diode that produces much stronger light compared to normal small LEDs.

2.DC MOTOR

A DC motor is an electromechanical device that converts direct current (DC) electrical energy into rotational mechanical motion. It is commonly used in automation systems, robotics, fans, pumps, and various motor-driven applications to provide controlled movement and power.

3.Potentiometer PCB

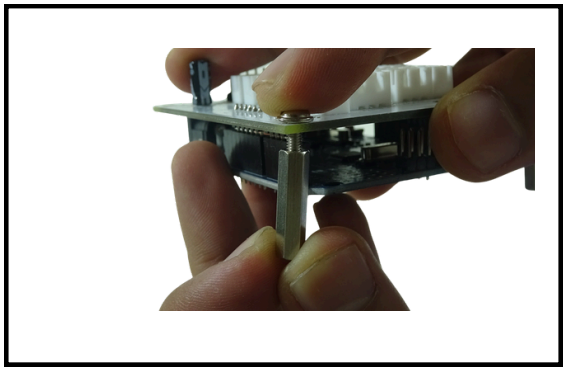
A potentiometer is a variable resistor that allows the adjustment of electrical resistance within a circuit. It is commonly used for controlling voltage levels, adjusting signal strength, and providing user input in electronic devices such as audio systems, control panels, and instrumentation equipment.

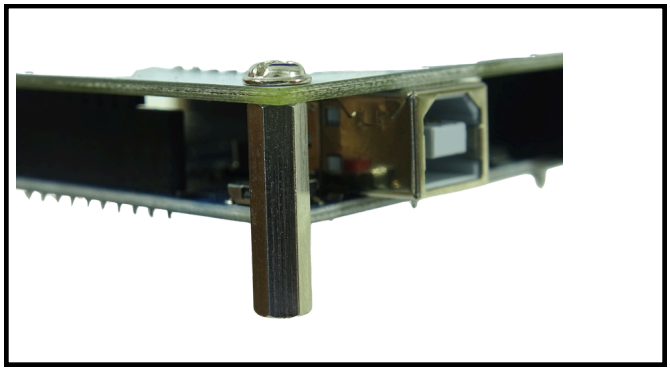
2 PIN RMC

Used to connect the 1-watt LEDs and the buzzer.

Procedure :

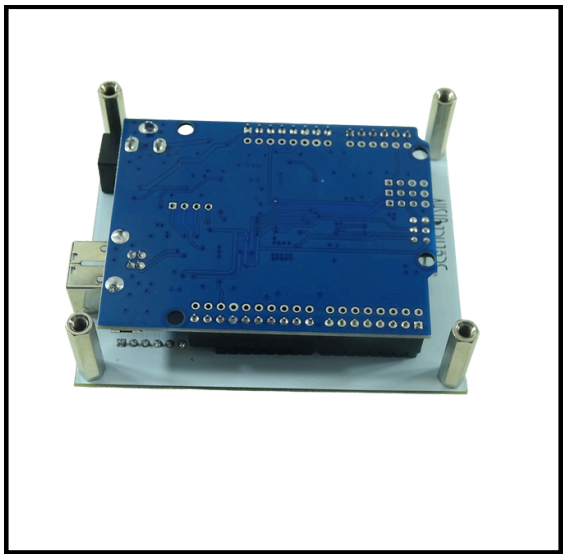
Step 1 : Fit the spacer to Arduino Shield as shown below using Spacer screw

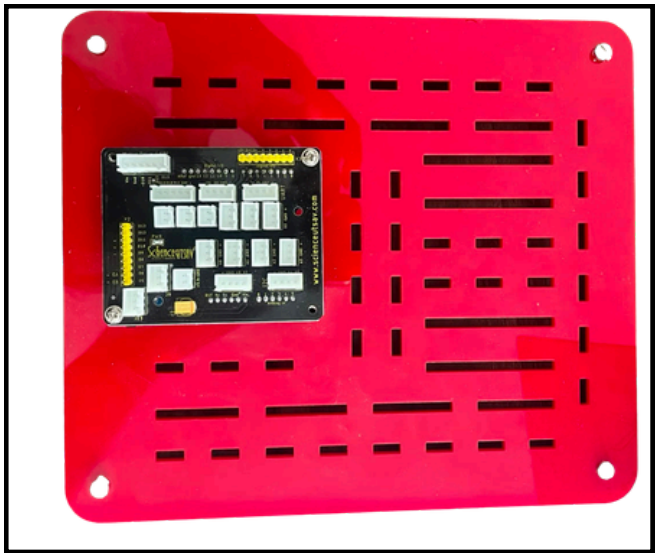




Step 2 : Repeat the step 1 for all 4 corners of Arduino Shield

Step 3 : Place the Arduino Shield to the base board as shown below

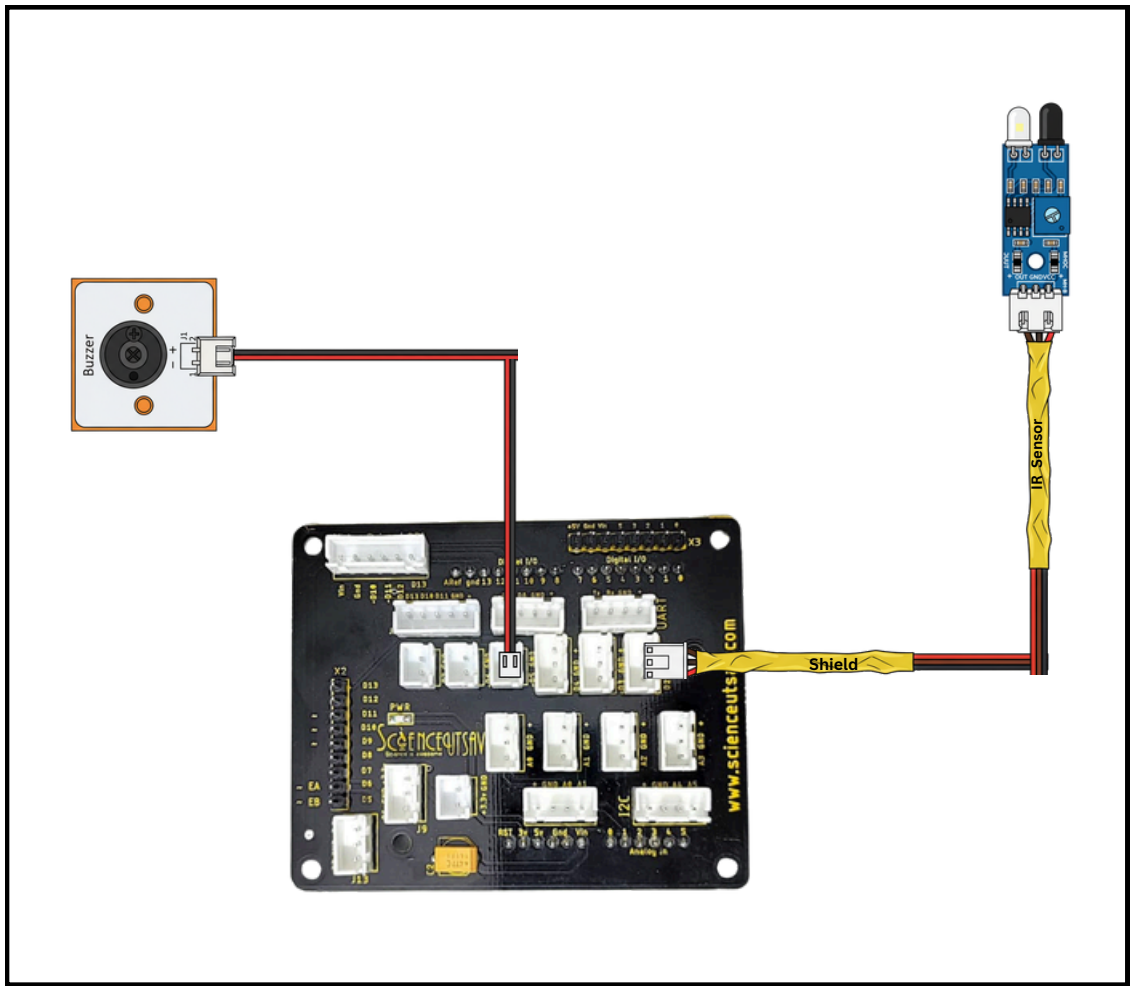




Step 4 : Fix it with the spacer screw from the bottom as shown below

BEGINNER PROJECT

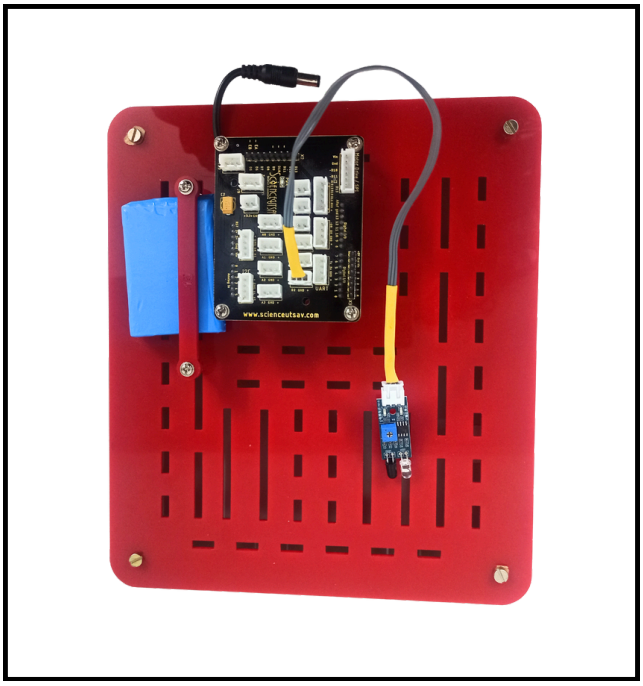
The circuit diagram of the project is given below



CONNECTION:

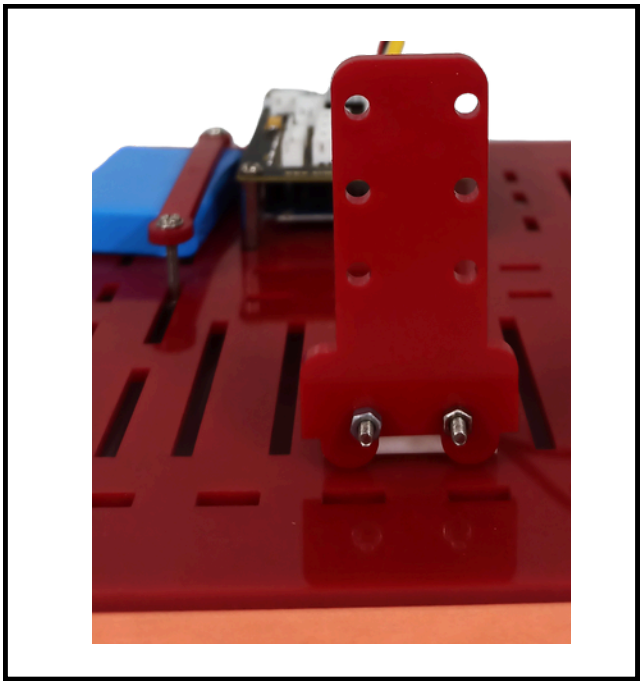
1. IR sensor- D2
2. Buzzer- D5

Step 1- Connect the IR sensor to the shield

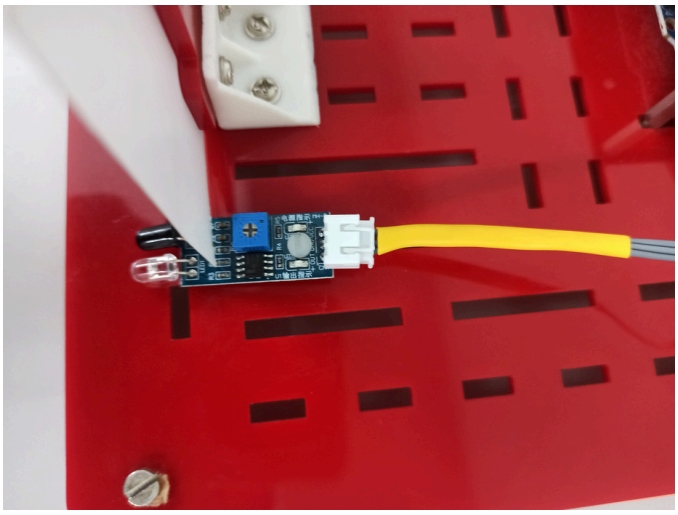


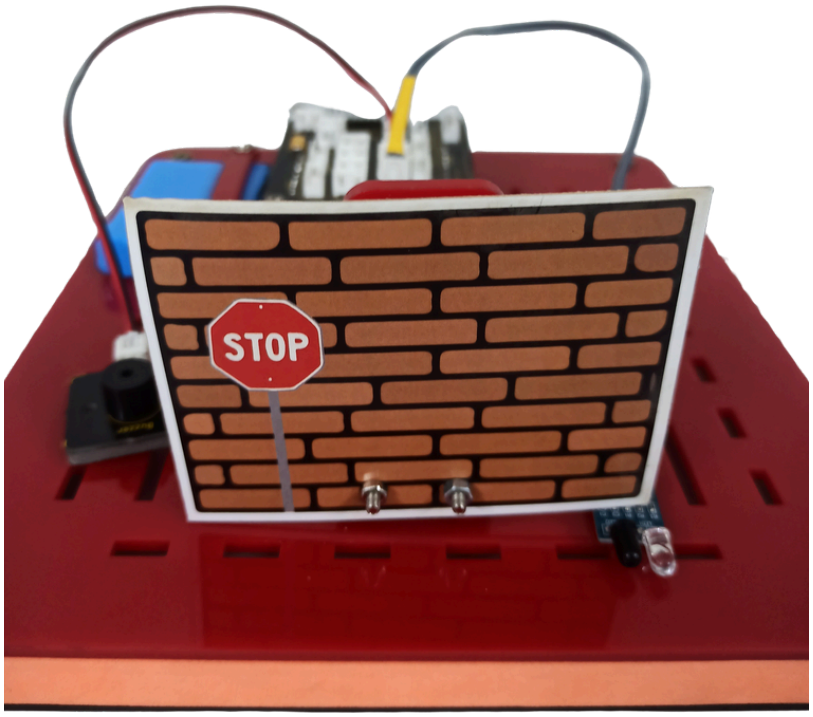
Step 2- Connect the buzzer as shown

Step 3- Fix the stand and attach the template to it



Step 4- Attach the IR sensor as shown



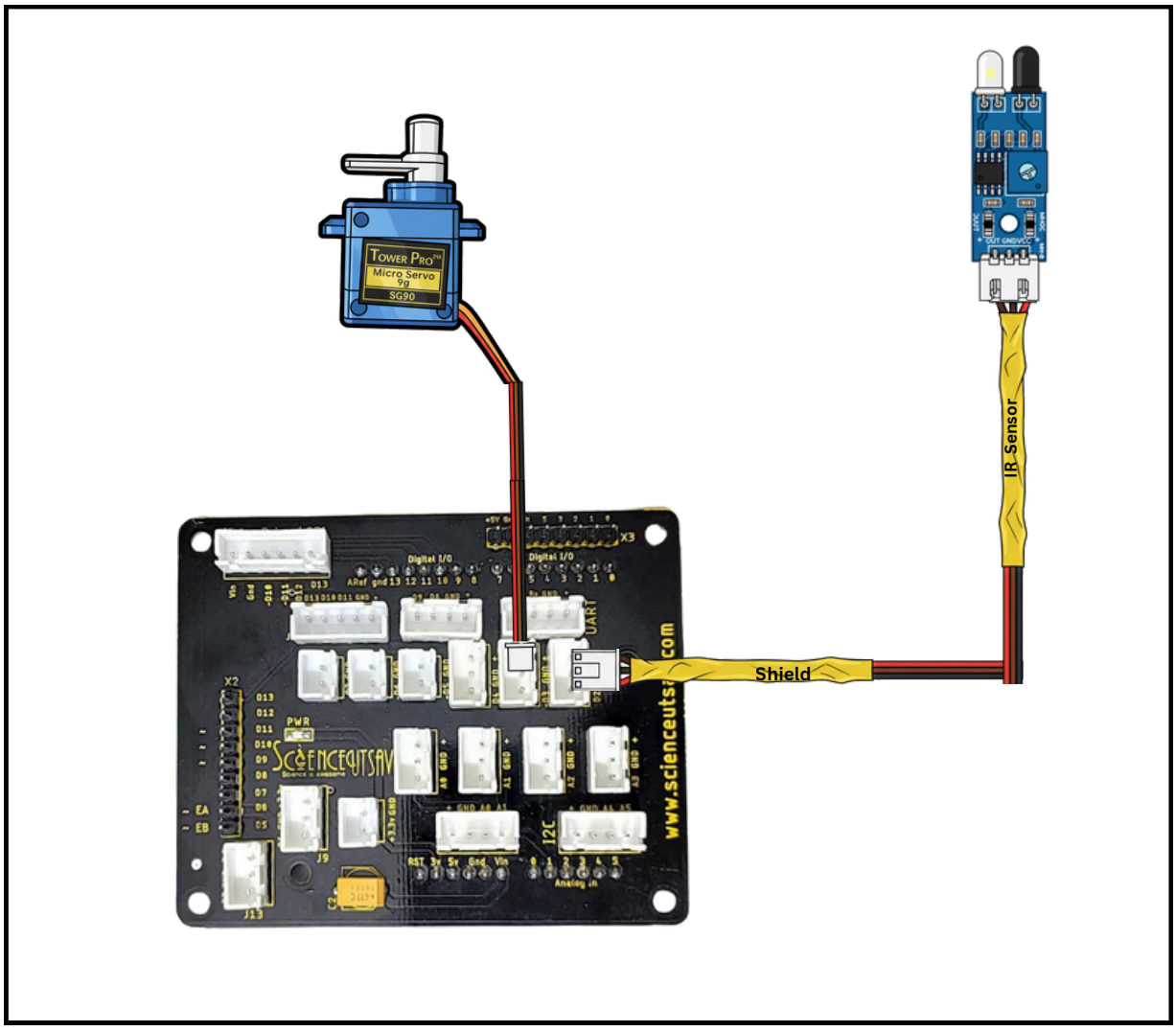


Code

```
// Session 5 Grade 6 Project 1
// Wall Parking Alert
// Object near wall = buzzer ON

const int irSensor = 2;
const int buzzer = 5;

const int IR_DETECTED = LOW;   // change to HIGH if opposite

void setup() {
  pinMode(irSensor, INPUT);
  pinMode(buzzer, OUTPUT);
  digitalWrite(buzzer, LOW);
}

void loop() {
  if (digitalRead(irSensor) == IR_DETECTED) {
    digitalWrite(buzzer, HIGH);
  } else {
    digitalWrite(buzzer, LOW);
  }
}
```

BEGINNER PROJECT

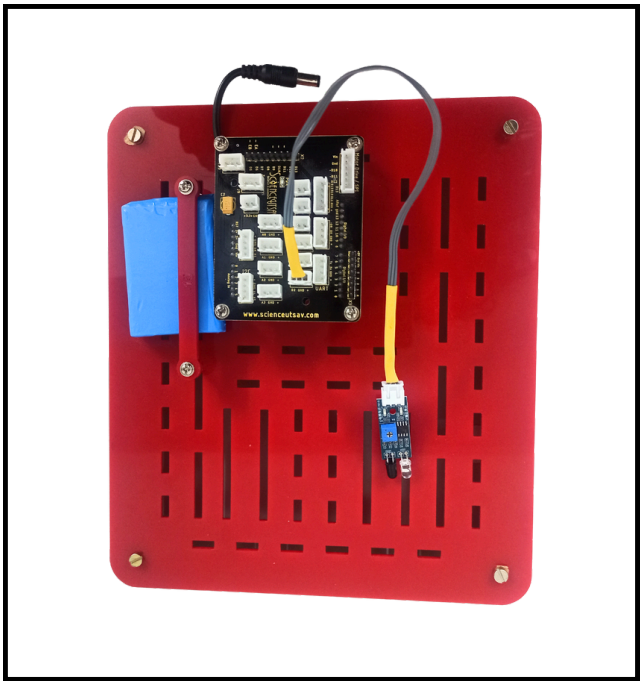
The circuit diagram of the project is given below



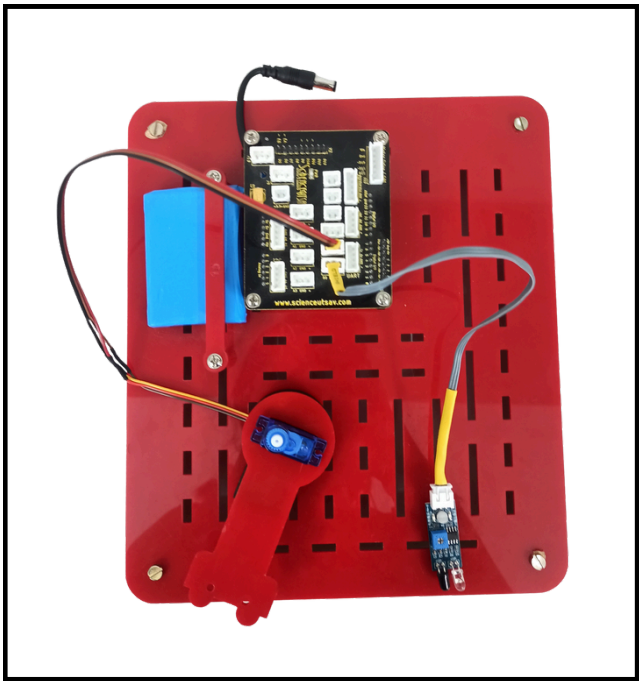
CONNECTION:

1. IR sensor- D2
2. Servomotor- D3

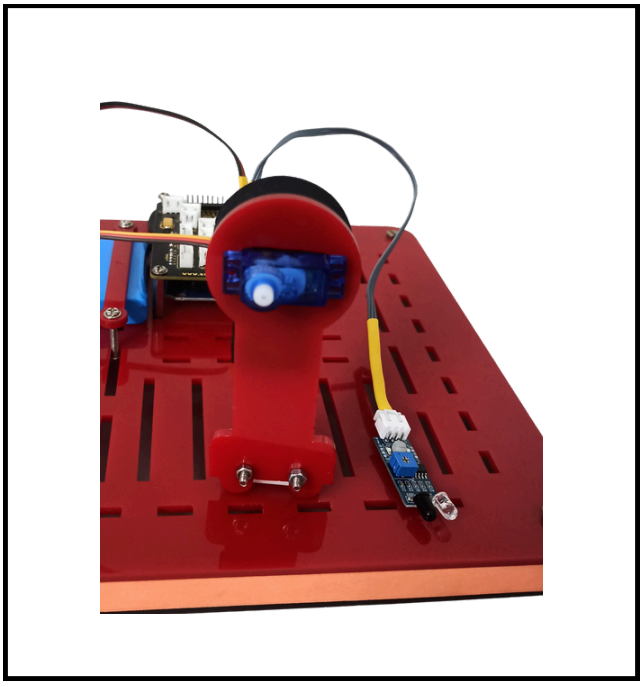
Step 1- Connect the IR sensor to the shield



Step 2- Connect the servomotor as shown

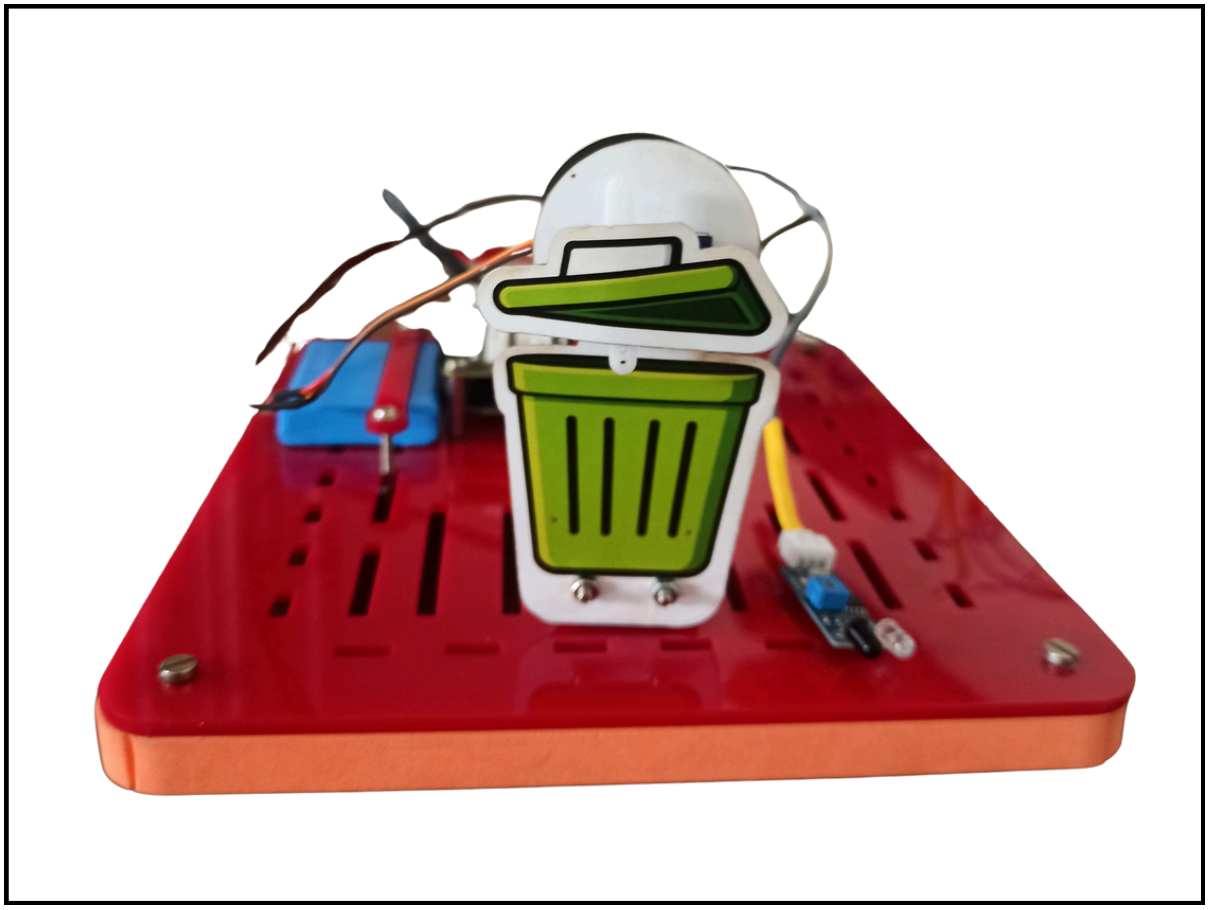


Step 3- Fix the stand to the board



Step 4- Attach the template to the stand and fix the shaft as shown

Step 5- Attach the other template to the shaft

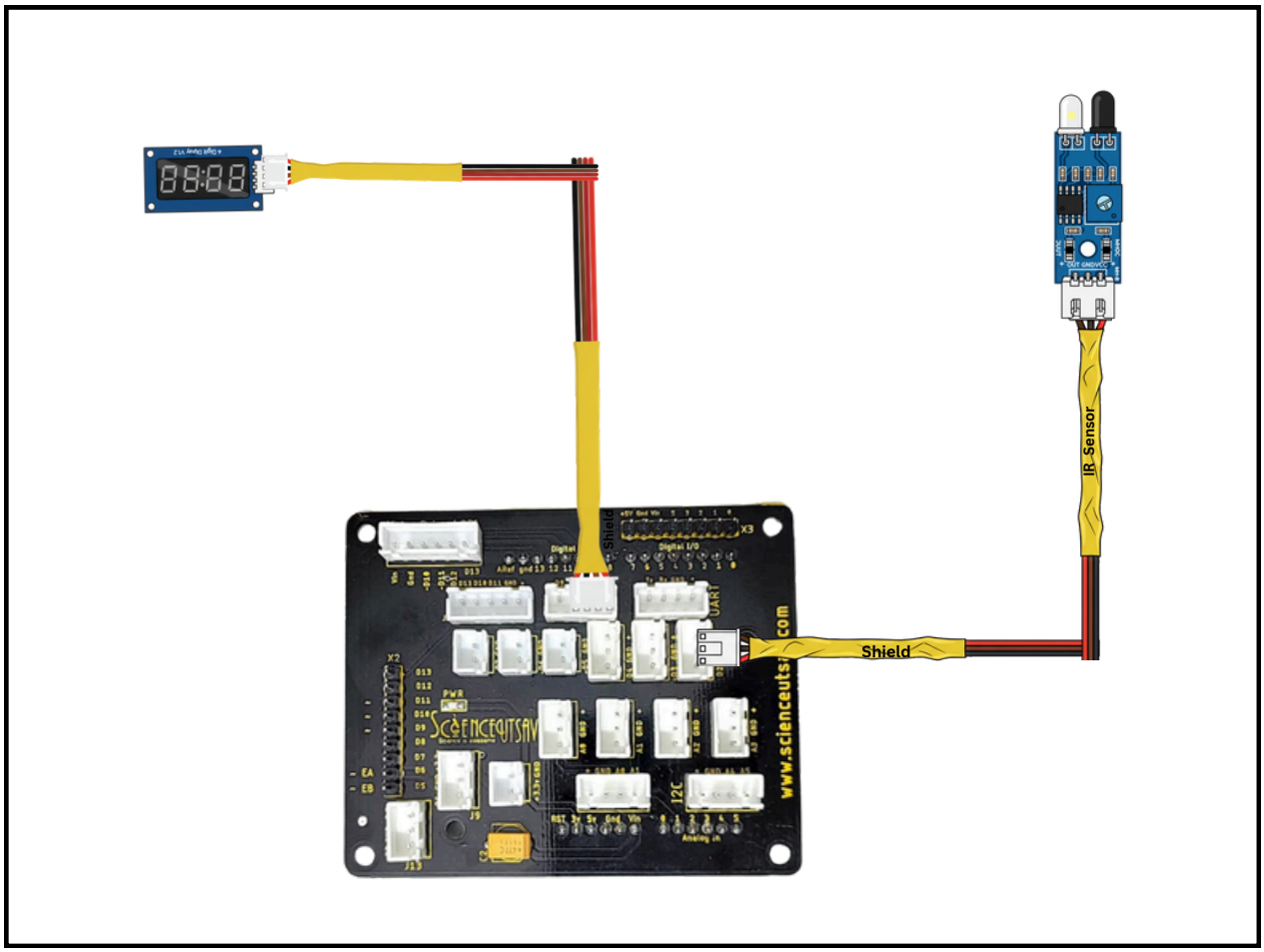


Code

```
// Session 5 Grade 6 Project 2
// Smart Trashcan Lid
// Person near dustbin = lid opens for 5 seconds

#include <Servo.h>

const int irSensor = 2;
const int servoPin = 3;

const int IR_DETECTED = LOW;   // change to HIGH if opposite

Servo lidServo;

void setup() {
  pinMode(irSensor, INPUT);
  lidServo.attach(servoPin);
  lidServo.write(0);          // lid closed
}

void loop() {
  if (digitalRead(irSensor) == IR_DETECTED) {
    lidServo.write(90);       // lid open
    delay(5000);
    lidServo.write(0);        // lid close
    delay(500);
  }
}
```

INTERMEDIATE PROJECT

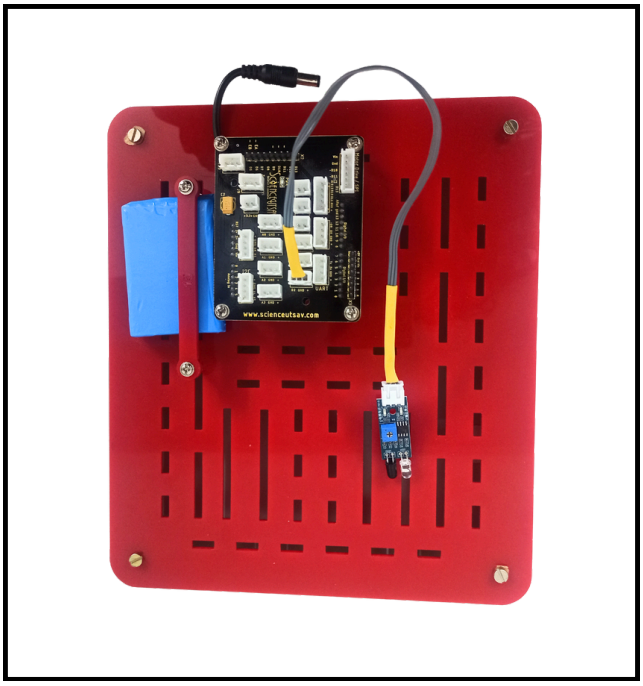
The circuit diagram of the project is given below



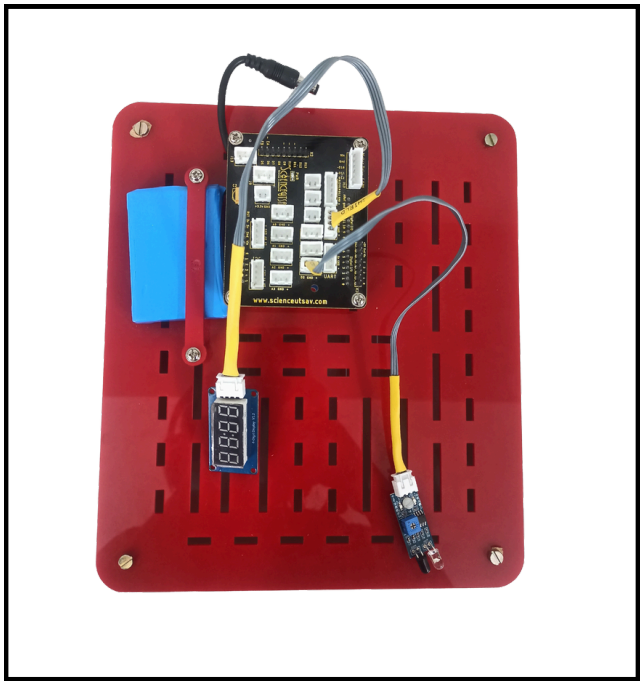
CONNECTION:

1. IR sensor- D2
2. 7-segment display- D8, D9

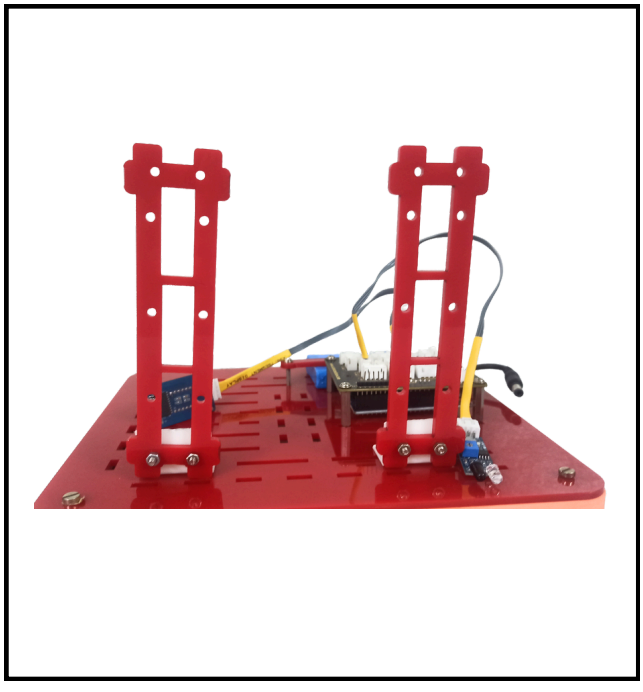
Step 1- Connect the IR sensor to the shield



Step 2- Connect the 7-segment dispay



Step 3- Fix the stands to the board as shown



Step 4- Attach the template using bolts

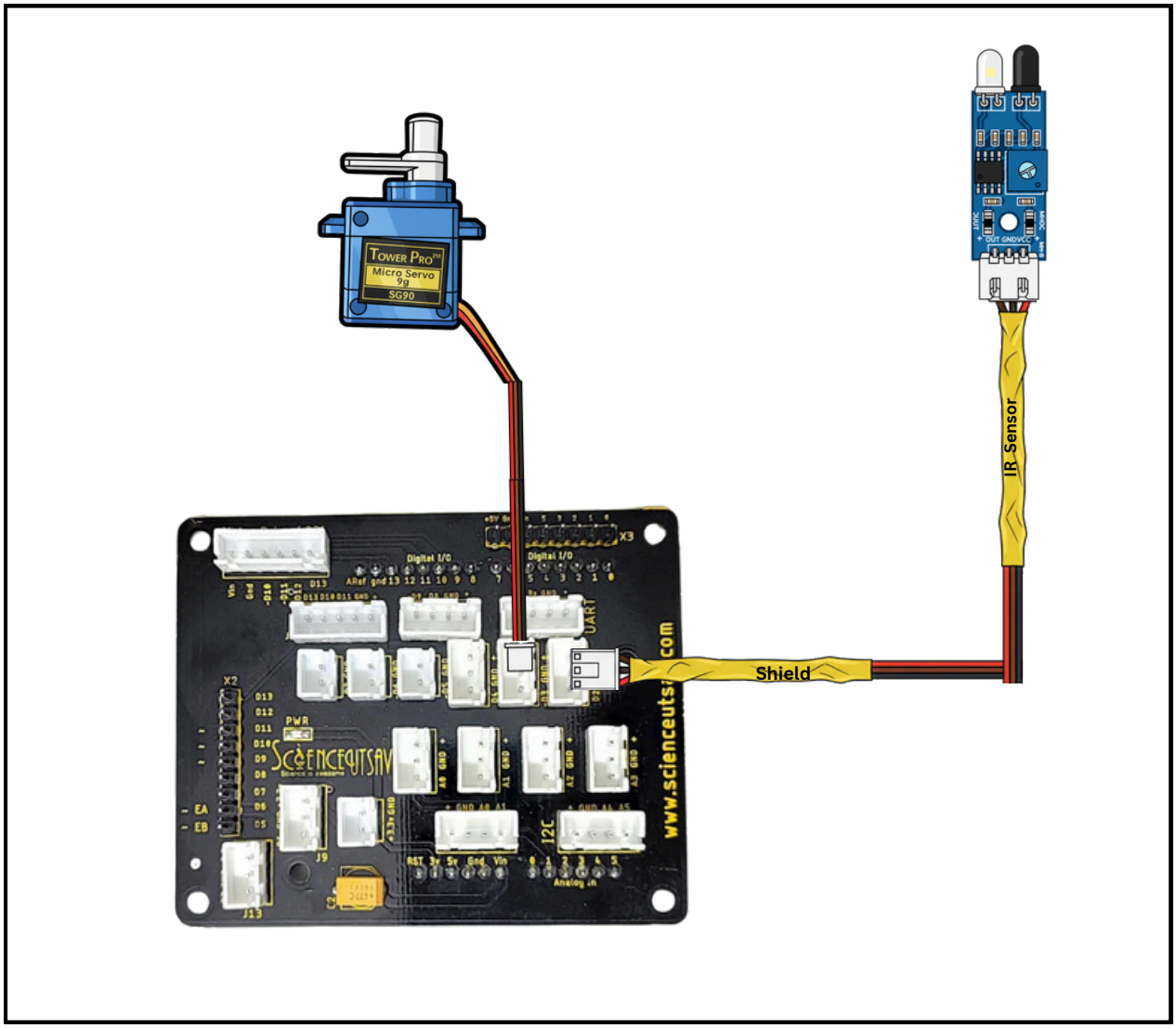
Step 5- Fix the display to the template as shown

Code

```
// Session 5 Grade 7 Project 1
// Arch Entry Counter
// Each person detected by IR = count increases on 7 segment

#include <TM1637Display.h>

const int irSensor = 2;
const int CLK = 9;
const int DIO = 8;

const int IR_DETECTED = LOW;   // change to HIGH if opposite

TM1637Display display(CLK, DIO);

int countValue = 0;
bool lastIR = HIGH;

void setup() {
  pinMode(irSensor, INPUT);
  display.setBrightness(7);
  display.showNumberDec(0, true);
}

void loop() {
  bool currentIR = digitalRead(irSensor);

  if (currentIR == IR_DETECTED && lastIR != IR_DETECTED) {
    countValue++;
    if (countValue > 9999) countValue = 0;
    display.showNumberDec(countValue, true);
    delay(300);
  }

  lastIR = currentIR;
}
```

INTERMEDIATE PROJECT

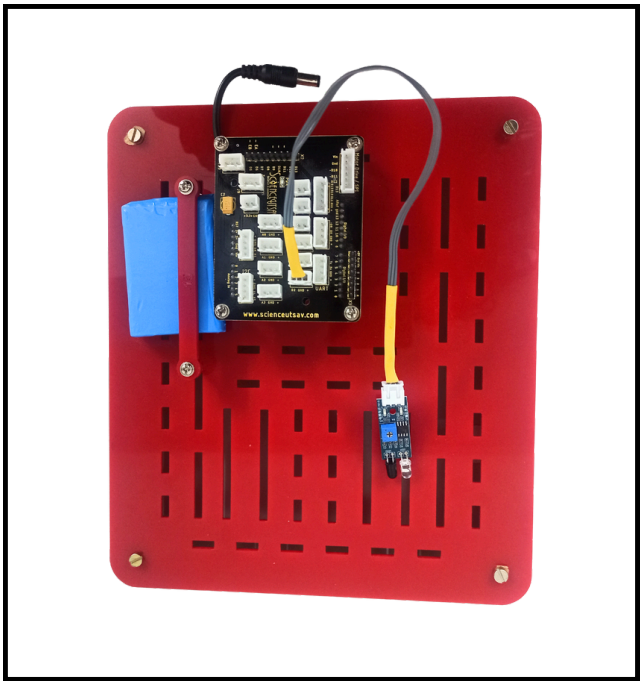
The circuit diagram of the project is given below



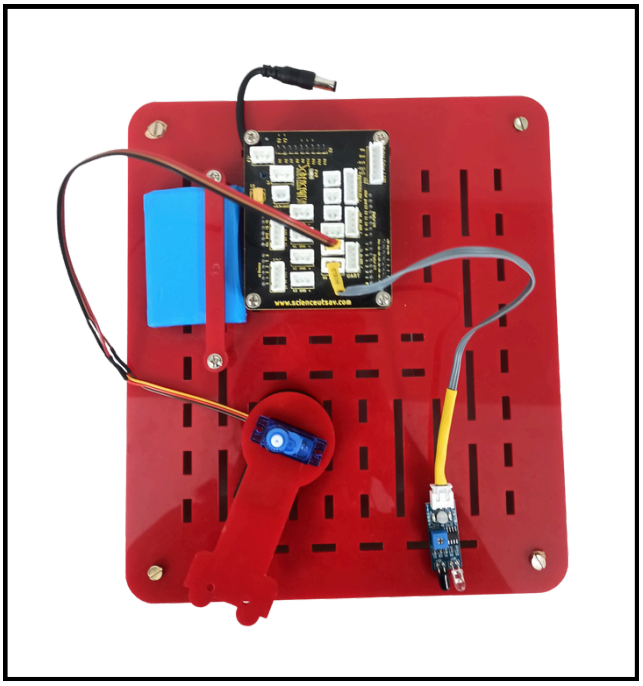
CONNECTION:

1. IR sensor- D2
2. Servomotor- D3

Step 1- Connect the IR sensor to the shield



Step 2- Connect the servomotor as shown



Step 3- Fix the stand to the board

Step 4- Attach the template to the servomotor as shown

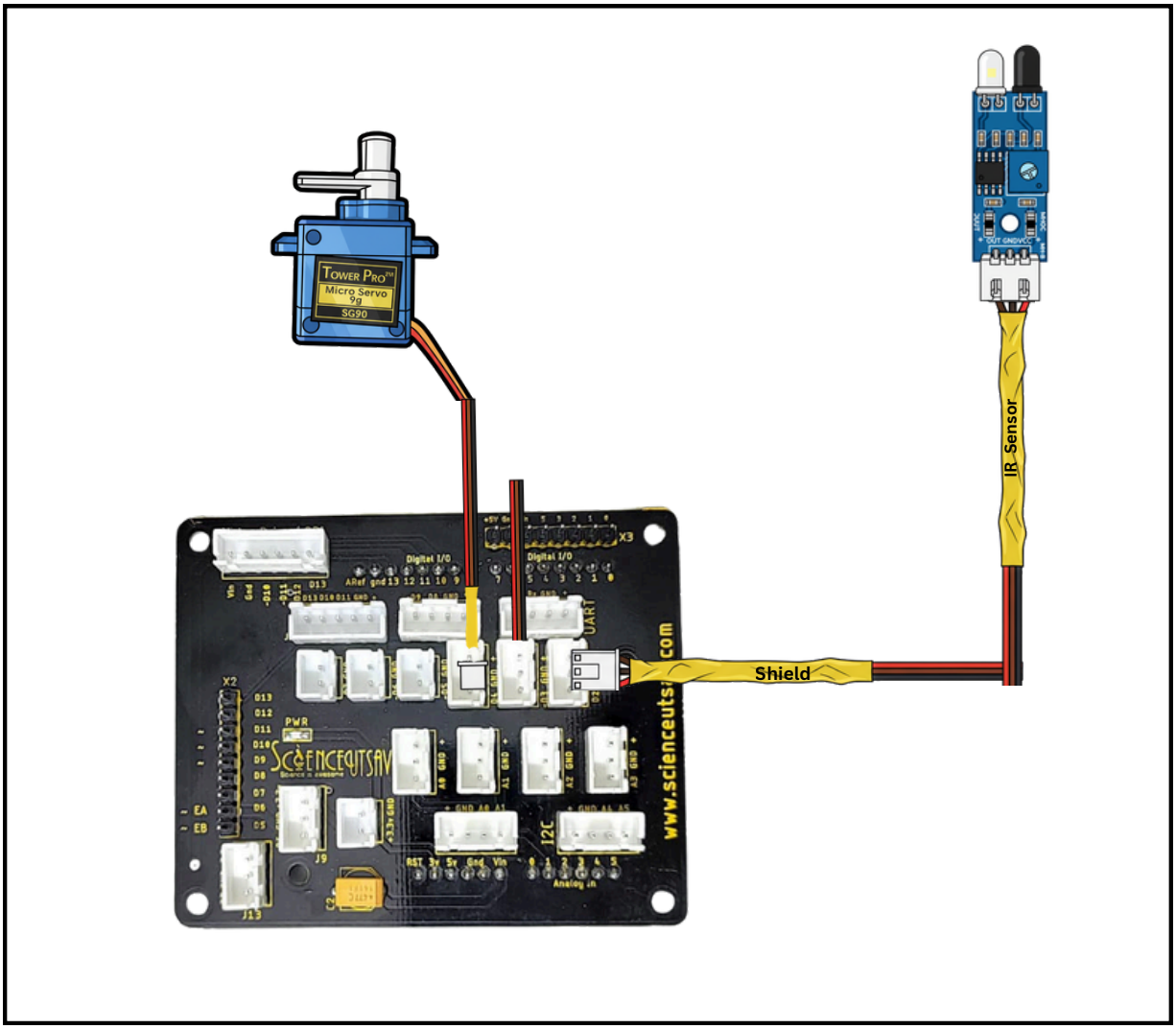
Step 5- Attach the other template to the board

Code

```
// Session 5 Grade 7 Project 2
// Automatic Servo Gate
// Person near gate = gate opens for 5 seconds

#include <Servo.h>

const int irSensor = 2;
const int servoPin = 3;

const int IR_DETECTED = LOW;   // change to HIGH if opposite

Servo gateServo;

void setup() {
  pinMode(irSensor, INPUT);
  gateServo.attach(servoPin);
  gateServo.write(0);         // gate closed
}

void loop() {
  if (digitalRead(irSensor) == IR_DETECTED) {
    gateServo.write(90);       // gate open
    delay(5000);
    gateServo.write(0);        // gate closed
    delay(500);
  }
}
```

ADVANCE PROJECT

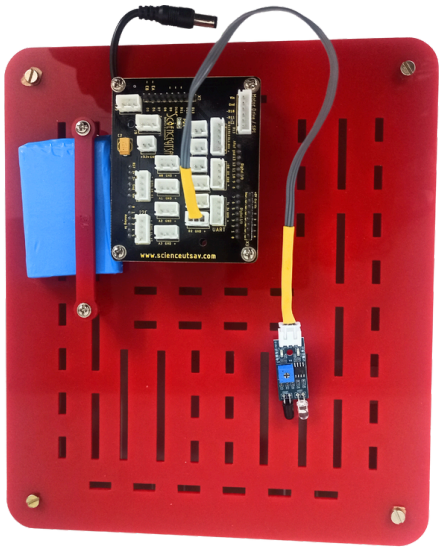
The circuit diagram of the project is given below



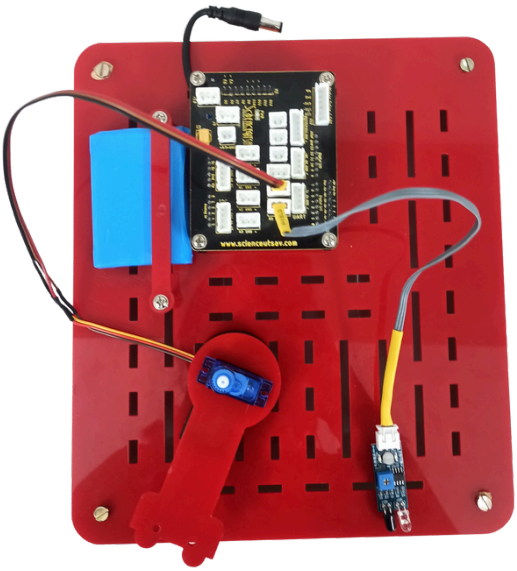
CONNECTION:

1. IR sensor- D2
2. Servomotor- D3

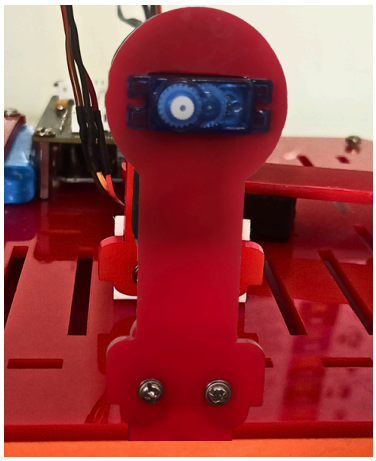
Step 1- Connect the IR sensor to the shield



Step 2- Connect the servomotor as shown



Step 3- Fix the stand to the board



Step 4- Attach the template to the stand as shown

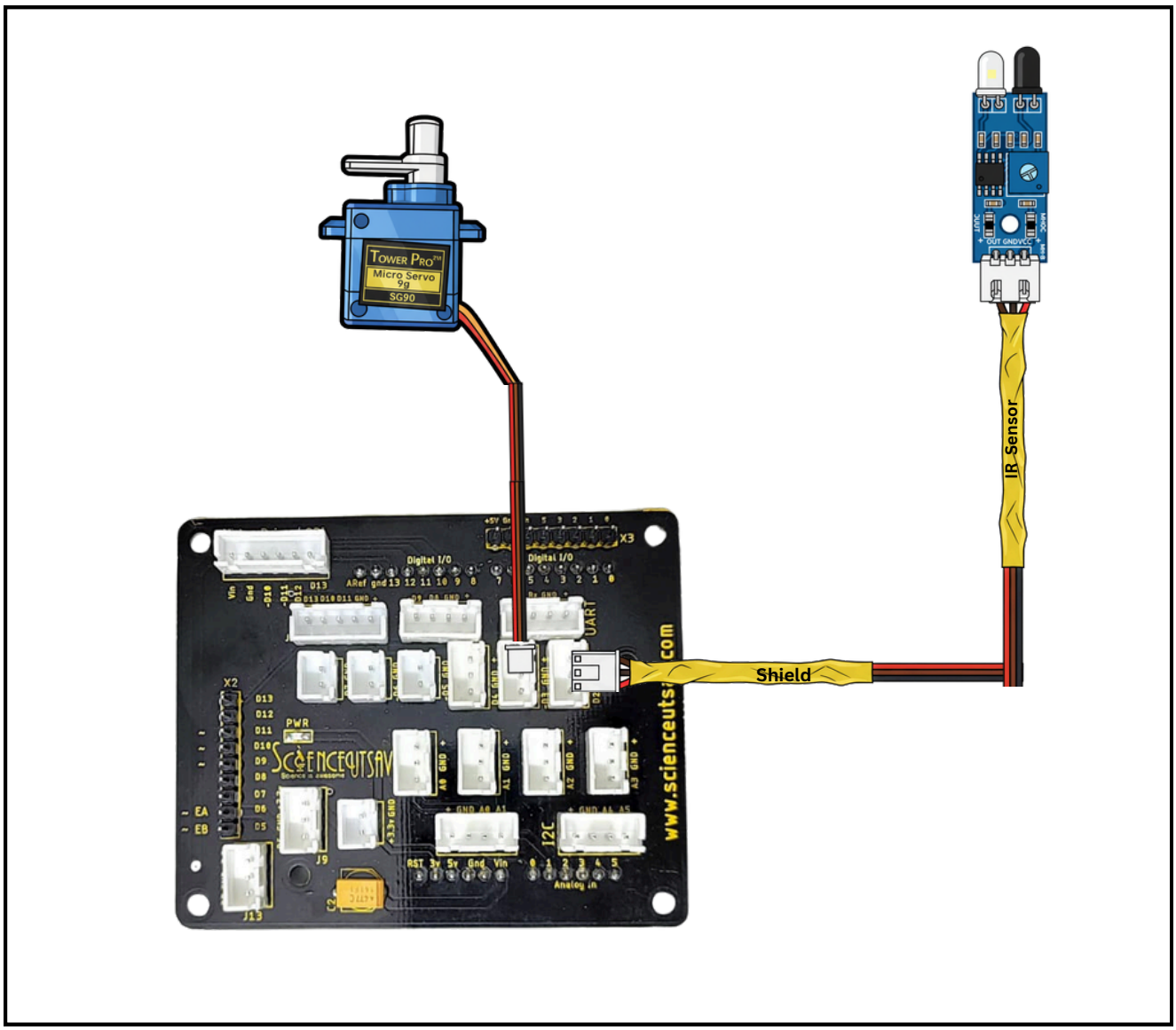
Step 5- Fix the other template to the servomotor

Code

```
// Session 5 Grade 8 Project 1
// Crocodile Mouth
// Hand near crocodile = mouth opens and closes

#include <Servo.h>

const int irSensor = 2;
const int servoPin = 4;

const int IR_DETECTED = LOW;   // change to HIGH if opposite

Servo mouthServo;

void setup() {
  pinMode(irSensor, INPUT);
  mouthServo.attach(servoPin);
  mouthServo.write(0);        // mouth closed
}

void loop() {
  if (digitalRead(irSensor) == IR_DETECTED) {
    mouthServo.write(80);      // mouth open
    delay(1000);
    mouthServo.write(0);       // mouth closed
    delay(500);
  }
}
```

ADVANCE PROJECT

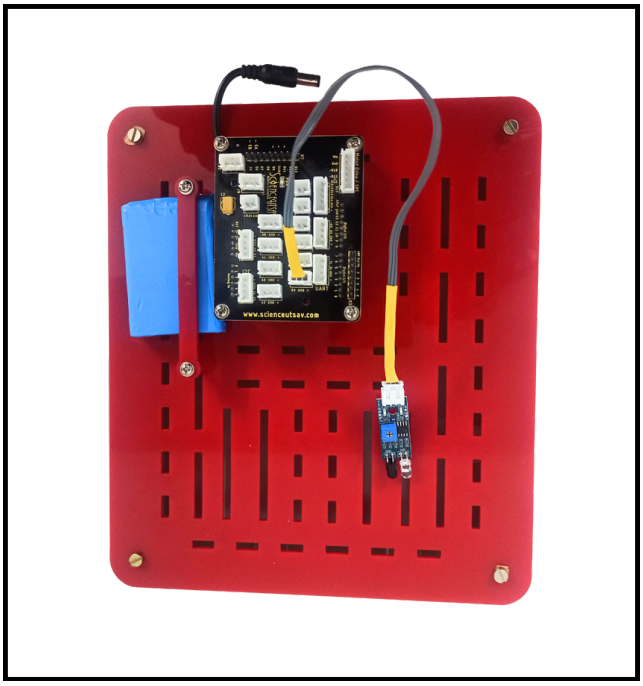
The circuit diagram of the project is given below



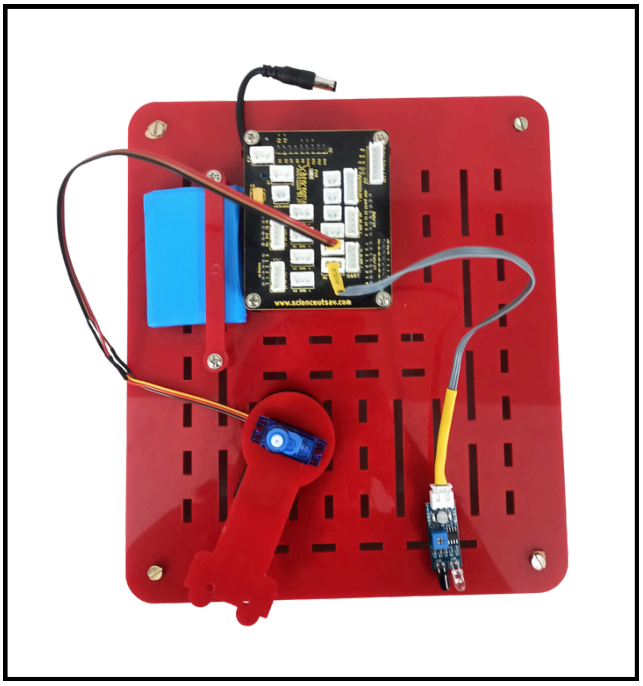
CONNECTION:

1. IR sensor- D2
2. Servomotor- D3

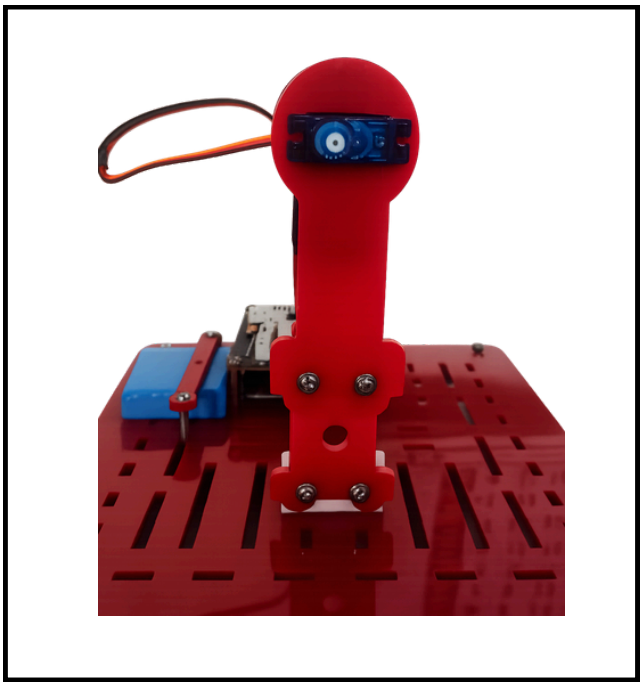
Step 1- Connect the IR sensor to the shield



Step 2- Connect the servomotor as shown

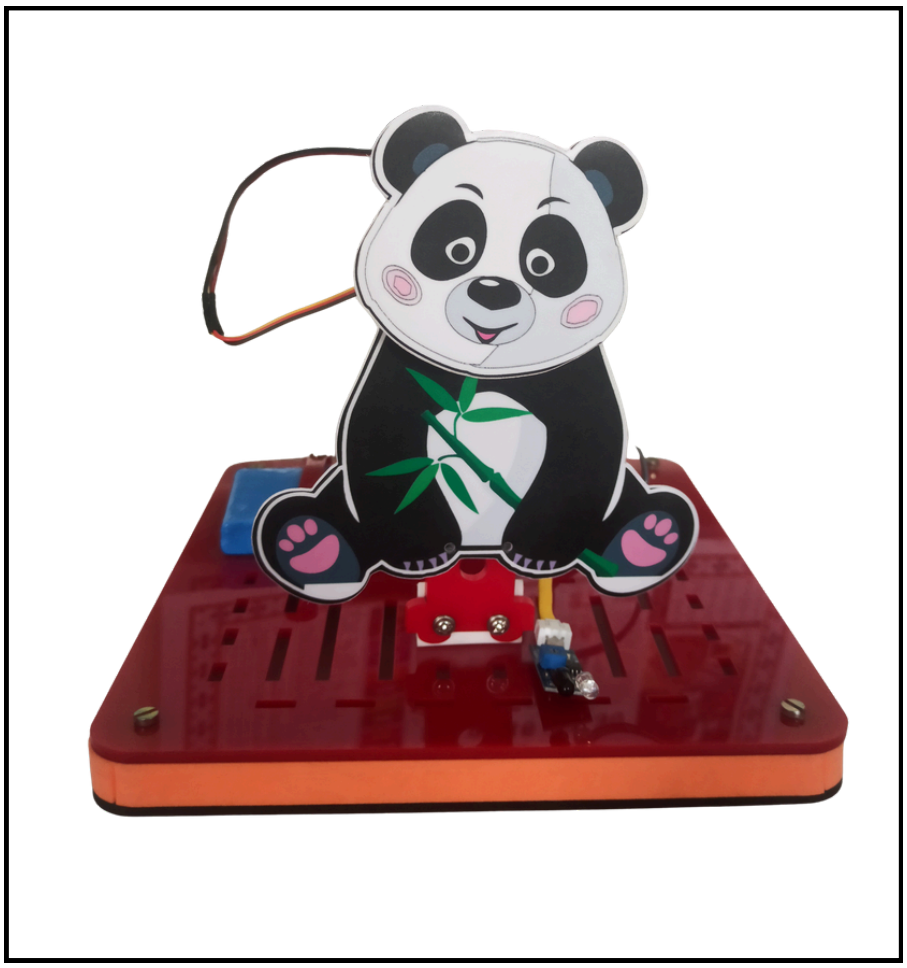


Step 3- Fix the stand to the board as shown



Step 4- Attach the template to the stand

Step 5- Fix the head to the motor as shown

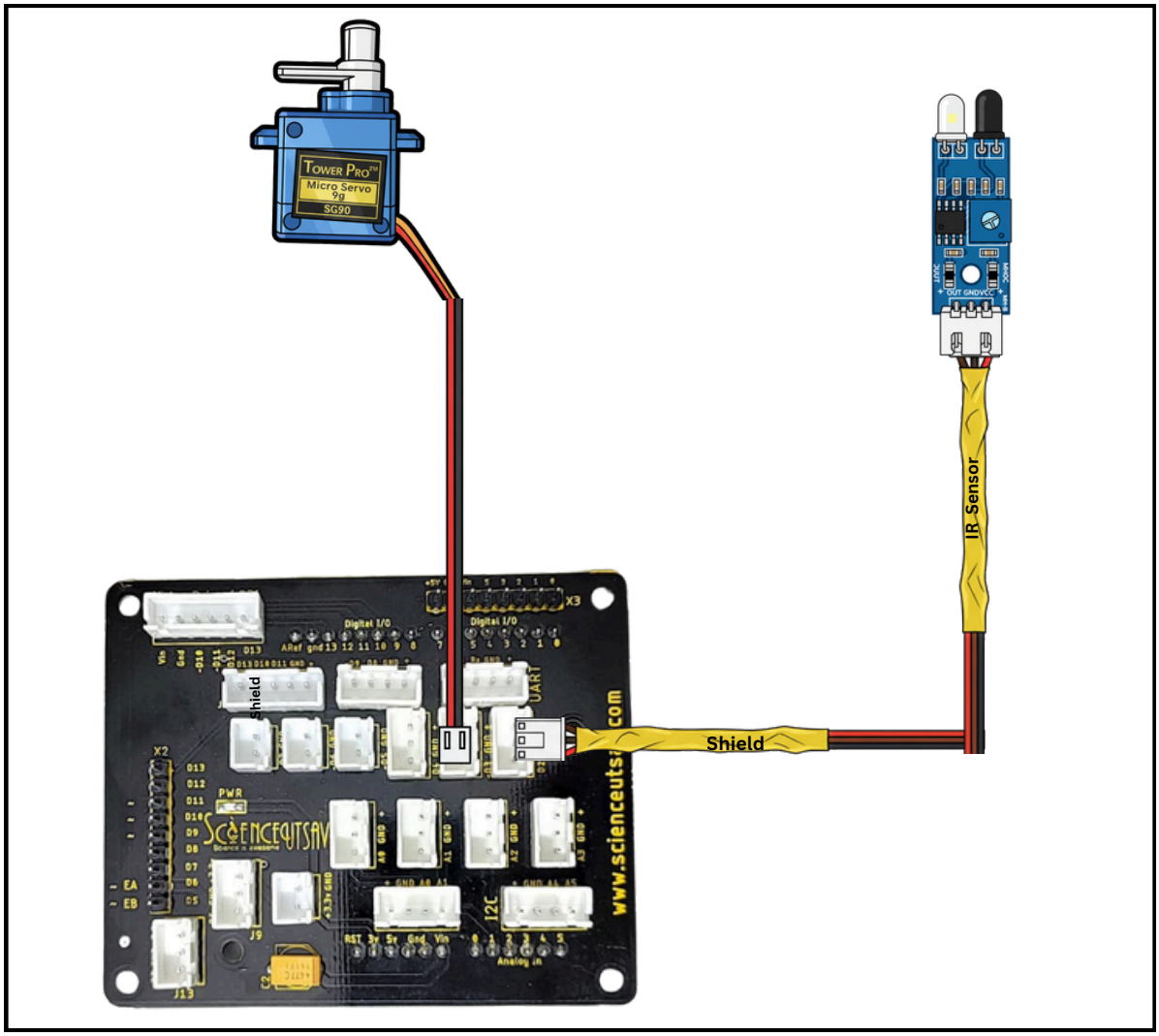


Code

```
// Session 5 Grade 8 Project 2
// Panda Face Movement
// Person near panda = face tilts

#include <Servo.h>

const int irSensor = 2;
const int servoPin = 3;

const int IR_DETECTED = LOW;   // change to HIGH if opposite

Servo faceServo;

void setup() {
  pinMode(irSensor, INPUT);
  faceServo.attach(servoPin);
  faceServo.write(90);        // center position
}

void loop() {
  if (digitalRead(irSensor) == IR_DETECTED) {
    faceServo.write(50);
    delay(500);
    faceServo.write(130);
    delay(500);
    faceServo.write(90);
    delay(500);
  }
}
```

EXPERT PROJECT

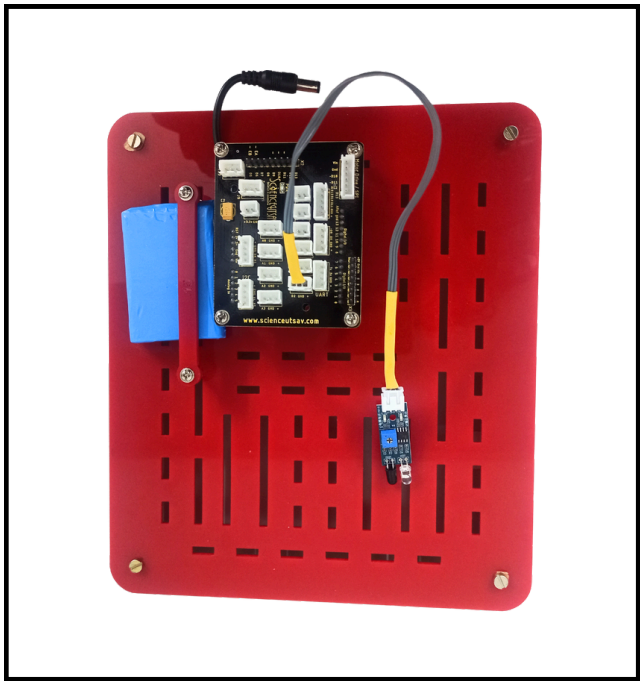
The circuit diagram of the project is given below



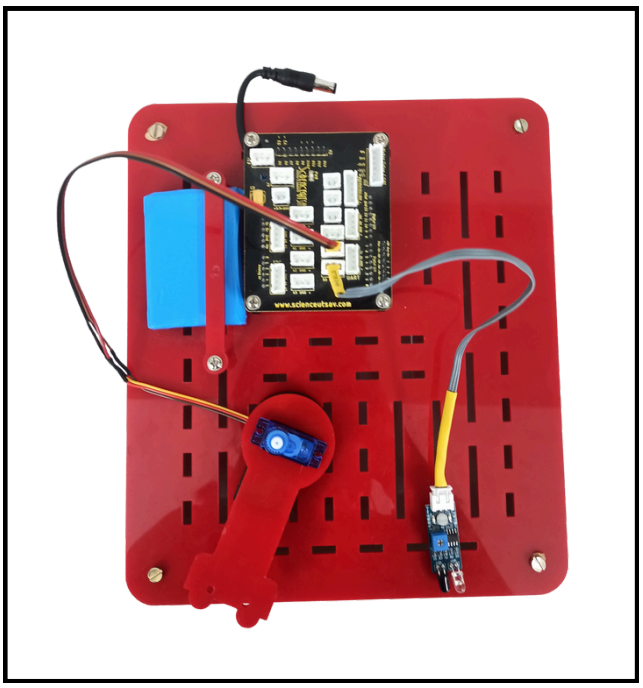
CONNECTION:

1. IR sensor- D2
2. Servomotor- D3

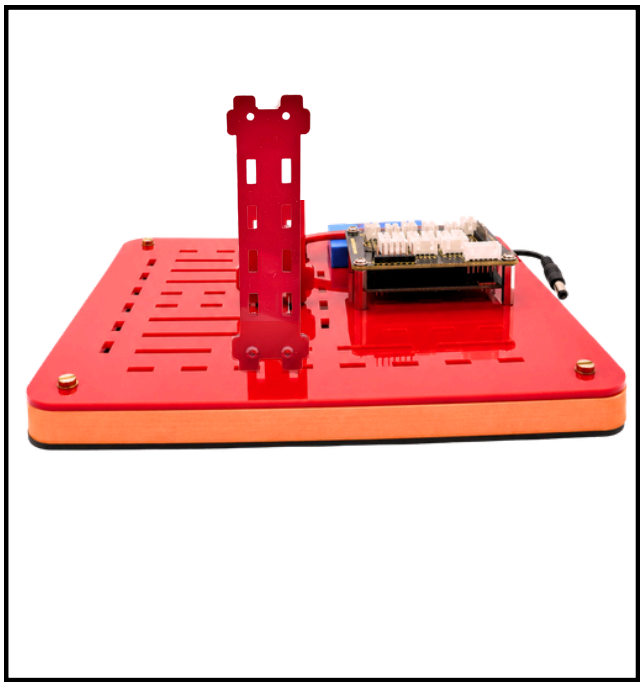
Step 1- Connect the IR sensor to the shield



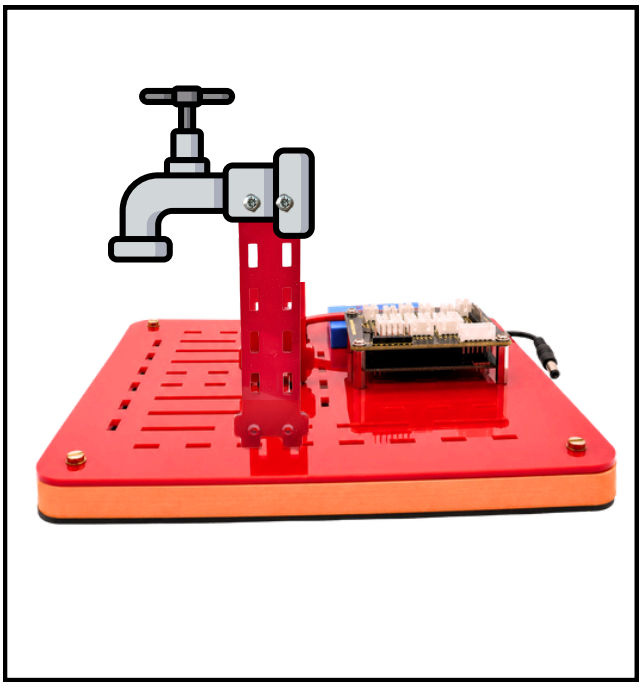
Step 2- Connect the servomotor as shown



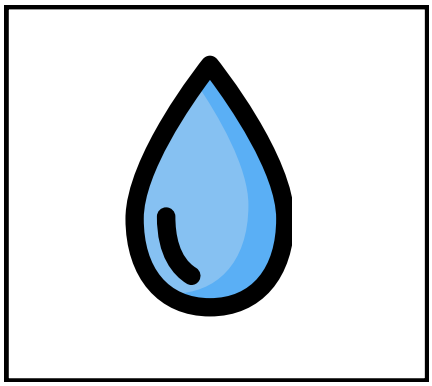
Step 3- Fix the stand to the board as shown

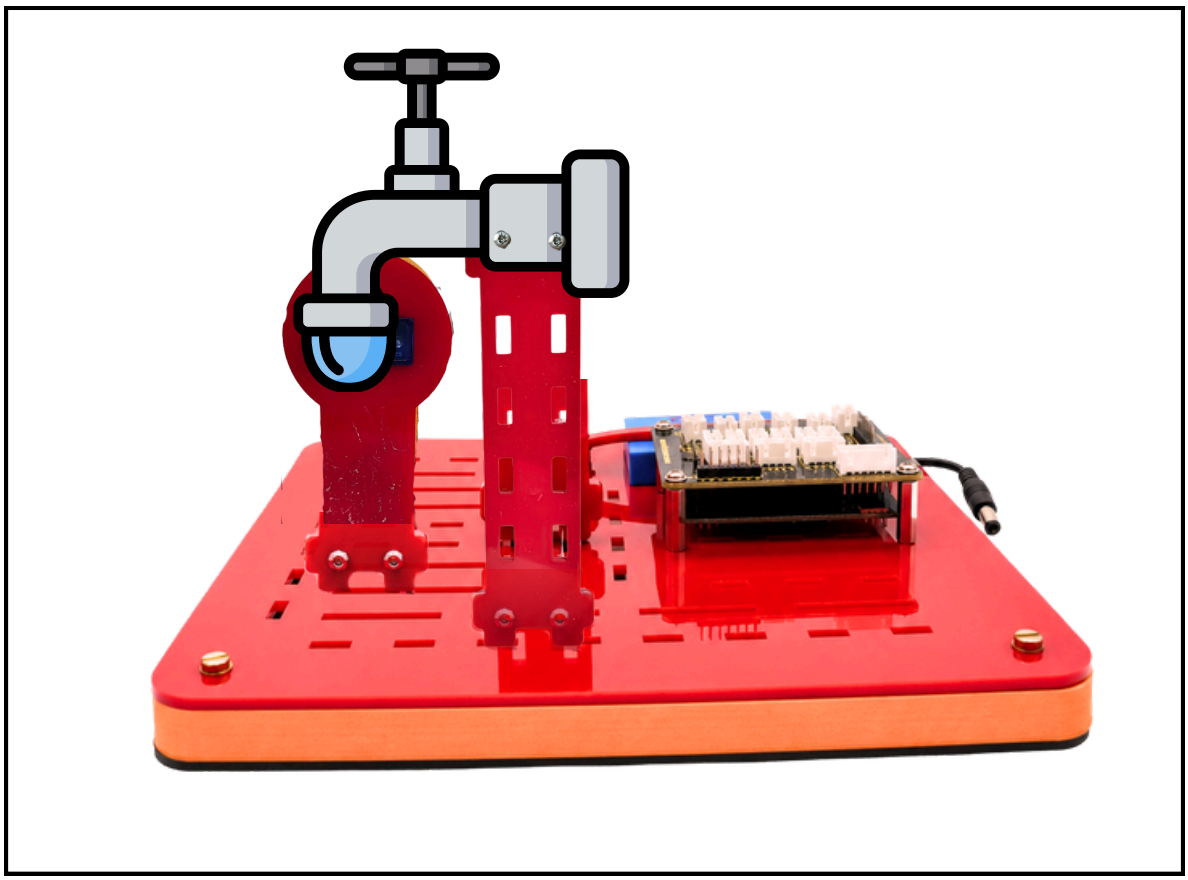


Step 4- Attach the template to the stand



Step 5- Attach the template to the motor



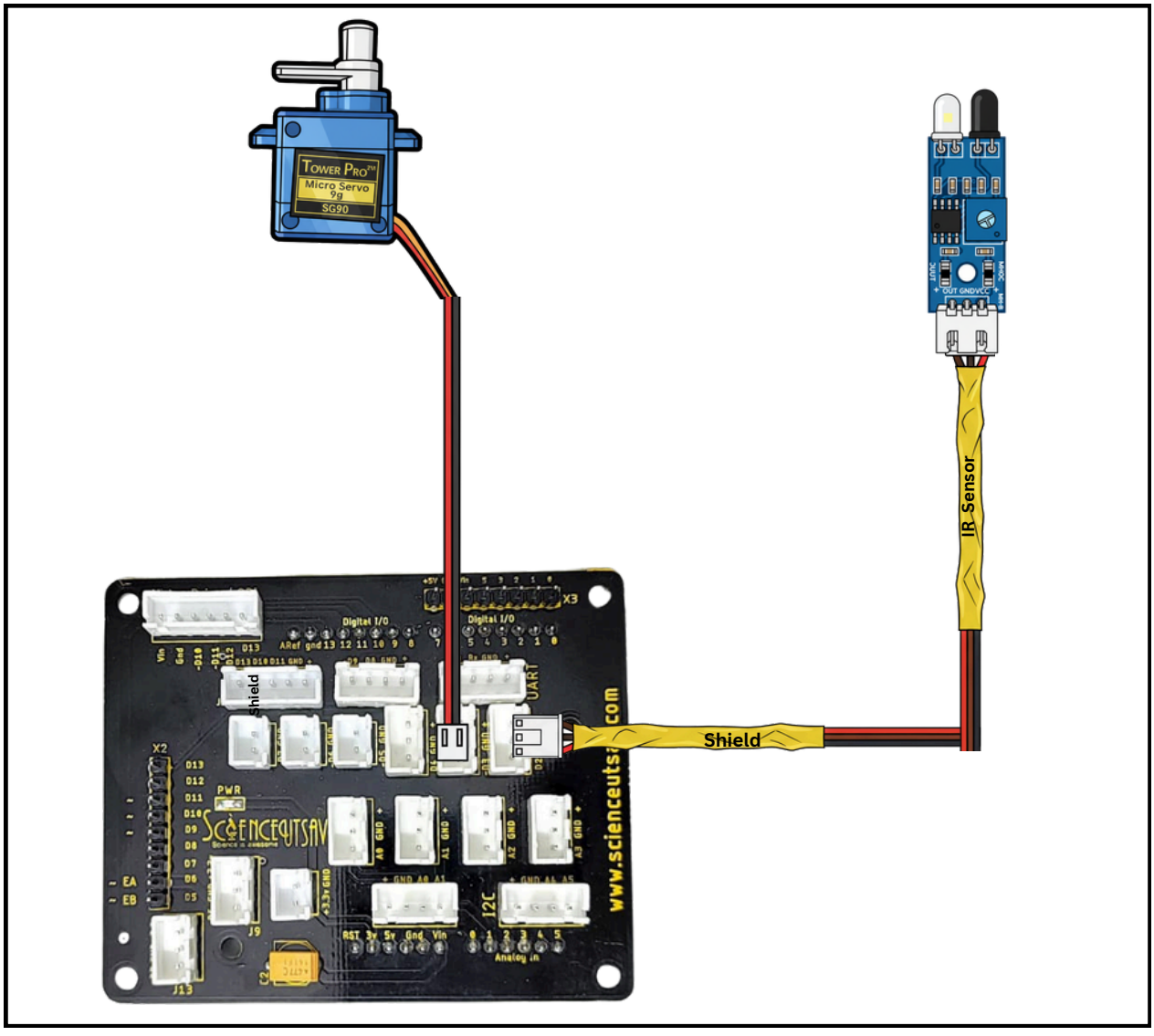


Code

```
// Session 5 Grade 9 Project 1
 // Smart Tap
 // Hand near tap = water droplet comes out

 #include <Servo.h>

 const int irSensor = 2;
 const int servoPin = 3;

 const int IR_DETECTED = LOW;  // change to HIGH if opposite

 Servo tapServo;

 void setup() {
 pinMode(irSensor, INPUT);
 tapServo.attach(servoPin);
 tapServo.write(0);          // tap off
 }

 void loop() {
 if (digitalRead(irSensor) == IR_DETECTED) {
   tapServo.write(90);       // water droplet out
 } else {
   tapServo.write(0);        // water droplet back
 }
 }
```

EXPERT PROJECT

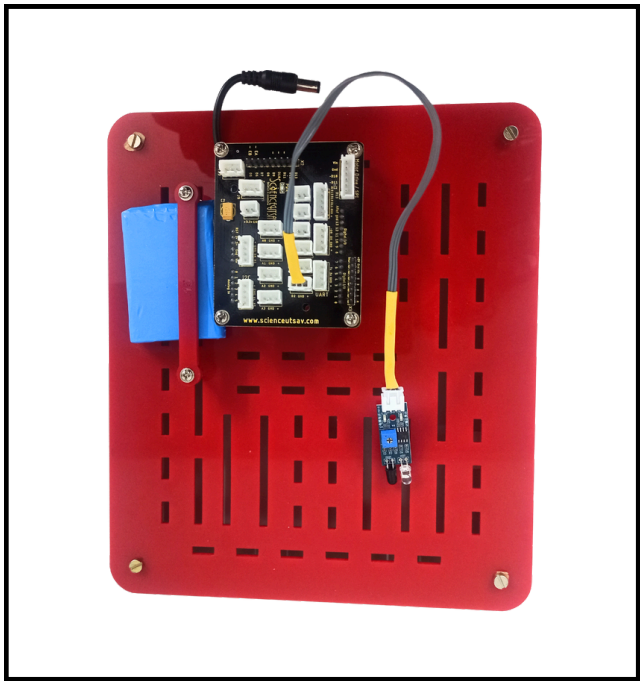
The circuit diagram of the project is given below



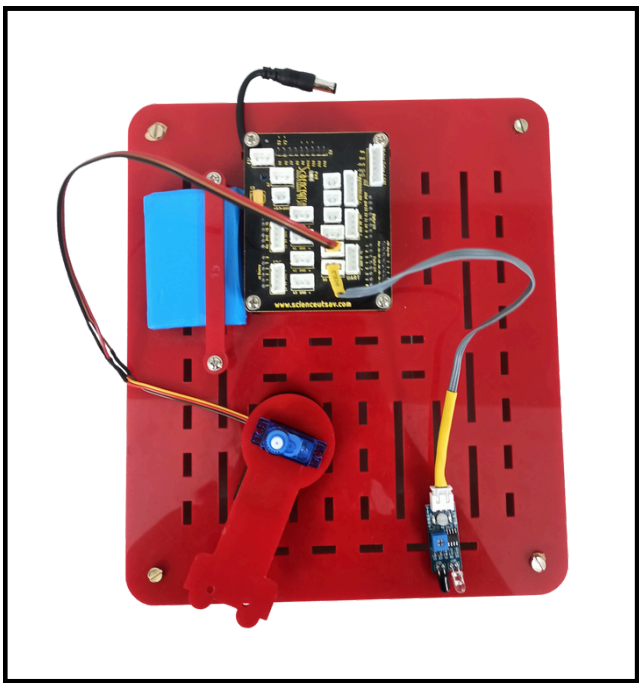
CONNECTION:

1. IR sensor- D2
2. Servomotor- D3

Step 1- Connect the IR sensor to the shield

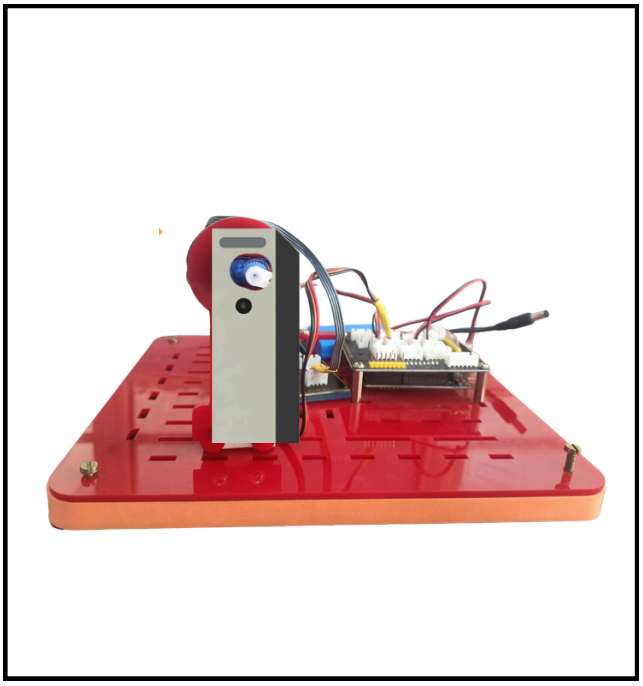


Step 2- Connect the servomotor as shown

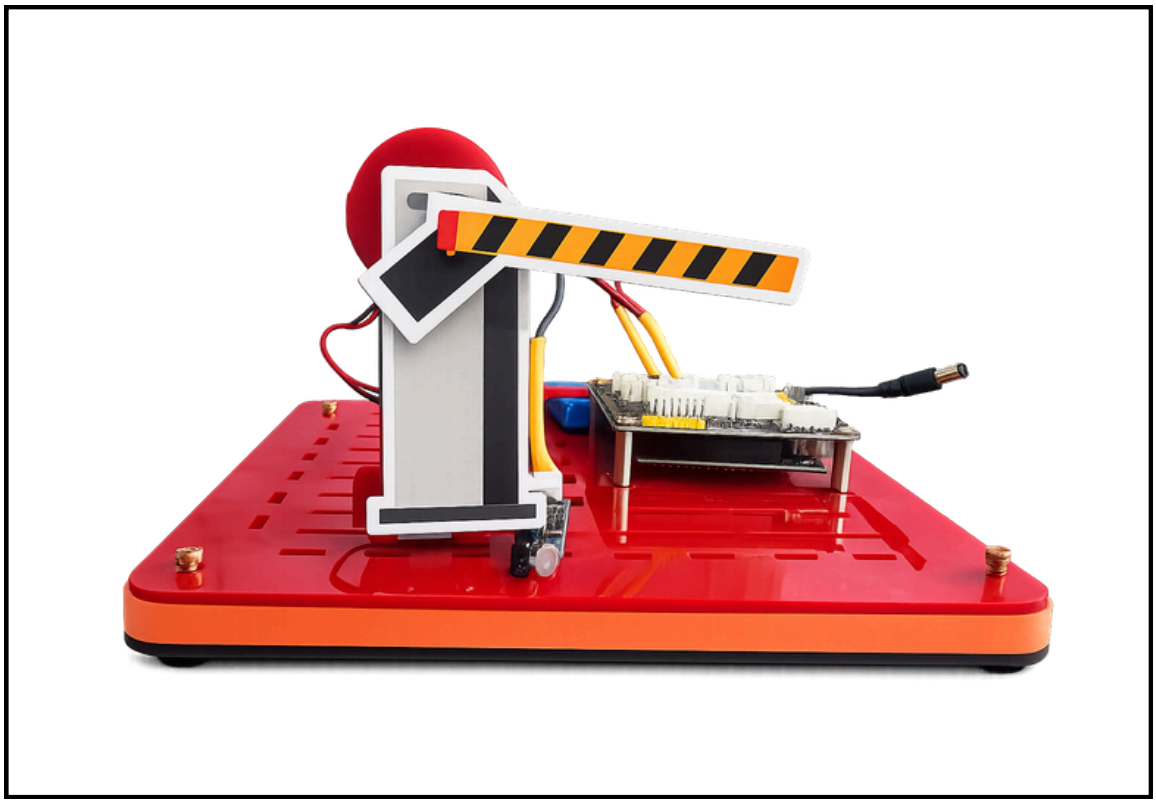


Step 3- Fix the stand to the board as shown

Step 4- Attach the template to the stand



Step 5- Attach the template to the stand



Code

```
// Session 5 Grade 9 Project 2
 // IR Servo Gate
 // IR detected = gate opens for 5 seconds

 #include <Servo.h>

 const int irSensor = 2;
 const int servoPin = 3;

 const int IR_DETECTED = LOW;  // change to HIGH if opposite

 Servo gateServo;

 void setup() {
 pinMode(irSensor, INPUT);
 gateServo.attach(servoPin);
 gateServo.write(0);        // gate closed
 }

 void loop() {
 if (digitalRead(irSensor) == IR_DETECTED) {
   gateServo.write(90);      // gate open
   delay(5000);
   gateServo.write(0);       // gate closed
   delay(500);
 }
 }
```